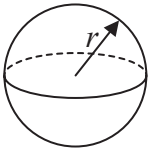
**14** Jan has some metal that she is going to make into solid metal spheres.

Each sphere will have a radius of 2.15 cm.

Jan has 1490 $cm^3$ of metal.

(a) Work out an estimate for the number of spheres that Jan can make.

Volume of sphere = $\frac{4}{3}\pi r^3$




Estimate1490≈1500, r=2.15≈2 and π≈3.

**1 mark**
Use sensible estimates

$\frac{4}{3} \times 3 \times 2^3 = 32\ cm^3$ per sphere.

Estimated number = 1500 ÷ 32 = 46.875.

**1 mark**
Metal ÷ sphere volume

About47 spheres (estimate).

**1 mark**
Estimate46–50

**(3)**

(b) If you calculate the number of spheres accurately, how do you think your answer to part (a) will change?
Give a reason for your answer.

The accurate number is less: the true radius and π are larger, so each sphere uses more metal; the actual amount of metal is also below1500$cm^3$.

**1 mark**
Less; larger divisors

**(1)**

**(Total for Question 14 is 4 marks)**